

20261008 Quiz

for (MATH102) Calculus II

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Problem 2.45. Consider the polar coordinate mapping

$$\mathbf{F}(r, \theta) = (r \cos \theta, r \sin \theta), \quad r \geq 0, \quad 0 \leq \theta \leq 2\pi.$$

Sketch a region in the (r, θ) -plane that corresponds to the upper half-plane $y > 0$.

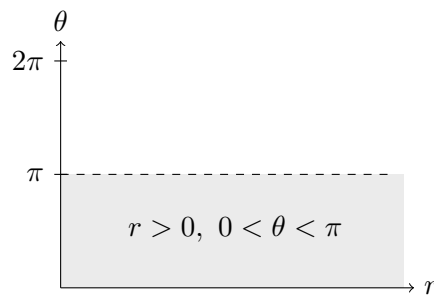
Solution. Since $y = r \sin \theta$, the condition $y > 0$ is equivalent, in the standard polar coordinate range, to

$$r > 0, \quad 0 < \theta < \pi.$$

Thus the required region is

$$\boxed{\{(r, \theta) \in \mathbb{R}^2 \mid r > 0, \ 0 < \theta < \pi\}}.$$

A sketch in the (r, θ) -plane is shown below.



□

Problem 2.46. Sketch the region in the (x, y) -plane that corresponds to the polar coordinate rectangle

$$1 \leq r \leq 2, \quad 0 \leq \theta \leq \pi.$$

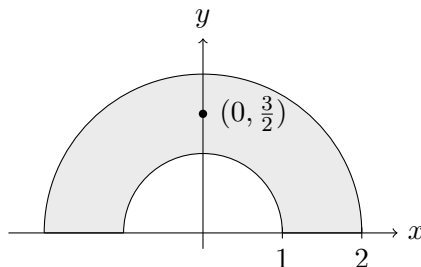
Find the polar coordinates of the point $(x, y) = (0, 1.5)$.

Solution. The condition $1 \leq r \leq 2$ places the points between the circles of radii 1 and 2, while $0 \leq \theta \leq \pi$ restricts them to the upper half-plane.

Hence the image is

$$\boxed{\{(x, y) \in \mathbb{R}^2 \mid y \geq 0, 1 \leq x^2 + y^2 \leq 4\}},$$

the closed upper half-annulus shown below.



For $(x, y) = (0, 1.5)$,

$$r = \sqrt{x^2 + y^2} = \frac{3}{2}, \quad \theta = \frac{\pi}{2}.$$

Therefore its polar coordinates are

$$\boxed{(r, \theta) = \left(\frac{3}{2}, \frac{\pi}{2}\right)}.$$

□

Problem 2.47. Use equations and inequalities to describe the following sets in polar coordinates.

- (a) The open unit disk $x^2 + y^2 < 1$.
- (b) The first quadrant $x > 0, y > 0$.

Solution. Using $x^2 + y^2 = r^2$ and the standard ranges $r \geq 0$ and $0 \leq \theta \leq 2\pi$,

- (a) The open unit disk is described by

$$\boxed{0 \leq r < 1, \quad 0 \leq \theta \leq 2\pi}.$$

- (b) In the first quadrant both $x = r \cos \theta$ and $y = r \sin \theta$ are positive. Thus

$$\boxed{r > 0, \quad 0 < \theta < \frac{\pi}{2}}.$$

□

Problem 2.49. Let $0 < b < a$ and consider the set of points in the plane whose polar coordinates r, θ satisfy

$$r = \frac{1}{a + b \cos \theta}.$$

- (a) Show that the x, y coordinates of the points satisfy

$$1 = a\sqrt{x^2 + y^2} + bx.$$

- (b) Show that the equation

$$1 = a\sqrt{x^2 + y^2} + bx$$

is the equation of an ellipse by completing a square and getting the equation into the form

$$1 = \frac{(x - \alpha)^2}{c^2} + \frac{(y - \beta)^2}{d^2}.$$

Solution.

(a) Multiplying the polar equation by $a + b \cos \theta$ gives

$$1 = ar + br \cos \theta.$$

Since

$$r = \sqrt{x^2 + y^2}, \quad r \cos \theta = x,$$

we obtain

$$1 = a\sqrt{x^2 + y^2} + bx.$$

(b) From the equation in part (a),

$$1 - bx = a\sqrt{x^2 + y^2}.$$

Squaring and rearranging gives

$$(a^2 - b^2)x^2 + 2bx + a^2y^2 = 1.$$

Since $0 < b < a$, we have $a^2 - b^2 > 0$. Completing the square in x gives

$$(a^2 - b^2) \left(x + \frac{b}{a^2 - b^2} \right)^2 + a^2y^2 = \frac{a^2}{a^2 - b^2}.$$

Therefore

$$1 = \frac{\left(x + \frac{b}{a^2 - b^2} \right)^2}{\frac{a^2}{(a^2 - b^2)^2}} + \frac{y^2}{\frac{1}{a^2 - b^2}}.$$

This is an ellipse with

$$\alpha = -\frac{b}{a^2 - b^2}, \quad \beta = 0, \quad c = \frac{a}{a^2 - b^2}, \quad d = \frac{1}{\sqrt{a^2 - b^2}}.$$

□

Problem 3.2. Let

$$f(x, y) = x^2 + 3y.$$

Find the linear approximation of $f(x, y)$ near $(2, 4)$, and use it to estimate $f(2.01, 4.03)$.

Solution. We have

$$f(2, 4) = 16, \quad f_x(x, y) = 2x, \quad f_y(x, y) = 3.$$

Thus

$$f_x(2, 4) = 4, \quad f_y(2, 4) = 3,$$

and the linear approximation at $(2, 4)$ is

$$L(x, y) = 16 + 4(x - 2) + 3(y - 4).$$

Using this approximation,

$$f(2.01, 4.03) \approx L(2.01, 4.03) = 16 + 4(0.01) + 3(0.03) = 16.13.$$

Thus the requested estimate is

$$f(2.01, 4.03) \approx 16.13.$$

□

Problem 3.3. Find the indicated partial derivatives of the functions.

(a) $f_x(x, y)$ and $f_y(2, 0)$ if

$$f(x, y) = e^{-x^2 - y^2}.$$

(b)

$$\frac{\partial}{\partial y} (xe^y + ye^x).$$

(c)

$$\frac{\partial}{\partial x} \left(\cos(xy) + \frac{\partial}{\partial y} \sin(xy) \right).$$

Solution.

(a) Differentiating with respect to x gives

$$\boxed{f_x(x, y) = -2xe^{-x^2 - y^2}}.$$

Also,

$$f_y(x, y) = -2ye^{-x^2 - y^2},$$

so

$$\boxed{f_y(2, 0) = 0}.$$

(b) Holding x fixed,

$$\boxed{\frac{\partial}{\partial y} (xe^y + ye^x) = xe^y + e^x}.$$

(c) First,

$$\frac{\partial}{\partial y} \sin(xy) = x \cos(xy).$$

Therefore

$$\begin{aligned} \frac{\partial}{\partial x} \left(\cos(xy) + \frac{\partial}{\partial y} \sin(xy) \right) &= \frac{\partial}{\partial x} (\cos(xy) + x \cos(xy)) \\ &= \boxed{\cos(xy) - y \sin(xy) - xy \sin(xy)}. \end{aligned}$$

□

Problem 3.6. Consider the linear function

$$\ell(x, y, z) = ax + by + cz.$$

Show that $\nabla \ell$ is constant. Show that when a, b, c are not all zero, $\nabla \ell$ is normal to the level set $\ell = 0$.

Solution. The partial derivatives are

$$\ell_x = a, \quad \ell_y = b, \quad \ell_z = c,$$

so

$$\boxed{\nabla \ell = (a, b, c)},$$

which is constant.

Now let $\mathbf{x} = (x, y, z)$ lie in the level set $\ell = 0$. Then

$$\nabla \ell \cdot \mathbf{x} = (a, b, c) \cdot (x, y, z) = ax + by + cz = 0.$$

Thus the level set $\ell = 0$ is the plane through the origin orthogonal to $\nabla \ell$. Since a, b, c are not all zero, $\nabla \ell \neq \mathbf{0}$, so $\nabla \ell$ is a normal vector to the level set.

□

Problem 3.11. Define functions $f, g: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \cos(x + y), \quad g(x, y) = \sin(2x - y).$$

Find the gradients ∇f and ∇g , and show that

$$f_x - f_y = 0, \quad g_x + 2g_y = 0.$$

Solution. For f ,

$$f_x(x, y) = -\sin(x + y), \quad f_y(x, y) = -\sin(x + y),$$

so

$$\boxed{\nabla f(x, y) = (-\sin(x + y), -\sin(x + y))}.$$

Hence

$$f_x - f_y = -\sin(x + y) + \sin(x + y) = 0.$$

For g ,

$$g_x(x, y) = 2 \cos(2x - y), \quad g_y(x, y) = -\cos(2x - y),$$

so

$$\boxed{\nabla g(x, y) = (2 \cos(2x - y), -\cos(2x - y))}.$$

Hence

$$g_x + 2g_y = 2 \cos(2x - y) - 2 \cos(2x - y) = 0.$$

□